Transport for London
Docklands Light Railway

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Engineering

EMC

DLR-ENG-STD-ES102 Issue B

MAYOR OF LONDON

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Contents

1. Purpose

This document provides requirements for a framework to ensure that the Electromagnetic Compatibility (EMC) aspects of equipment (i.e. Apparatus, Systems and Fixed Installations) within Docklands Light Railway (DLR) Limited are adequately addressed. Such arrangements are necessary to comply with both legal and internal approval requirements.

2. Application

This standard applies to all new works on the DLR and to alteration and upgrading work where specified in the contract or concession agreement.

The use of ‘shall’ in this standard indicates a mandatory requirement.

The use of ‘should’ in this standard indicates guidance or good practice but not a mandatory requirement.

3. References

References in this standard are made to latest editions unless specific editions are cited.

The list of references below and the references quoted in this standard are included for information only and should not be regarded as an exhaustive list.

Where any conflicts arise between published standards and this ES, compliance with this document shall take precedence except where a breach of law would arise.

DLR-ENG-STD-ES-201	Communications Systems
DLR-ENG-STD-ES-301	Signalling Principals
DLR-ENG-STD-ES-302	Automatic Train Control
DLR-ENG-STD-ES-601	Electrical Power Supplies
DLR-ENG-STD-ES-602	Building and Other Services, Mechanical and Electrical

DLR-ENG-STD-ES-604	Earthing Bonding and Corrosion Protection
DLR-ENG-STD-ES-605	Electrical Installation Standard
DLR-ENG-STD-ES-606	750V Traction Distribution Protection System
DLR-ENG-STD-ES-609	High Voltage and Traction Power Supply
DLR-ENG-STD-ES-708	Rolling Stock – Power Supply Interface
DLR-IMS-GENR-BCP-00009 (BCP-09)	Management of DLR Engineering & Maintenance Standards
S1222	LU Electromagnetic Compatibility
1-193 (S1193)	EMC with LU Signalling System Assets
S1196	LU Signalling And Signalling Control - Concept And Requirements
S1622	LU Glossary Of Terms And Abbreviations Used In Corporate Engineering Documents
S1691	LU Information
G222	LU Manual of EMC Best Practice
NR/L1/RSE/30040	EMC Strategy for Network Rail
NR/L2/RSE/30041	EMC Assurance Process for Network Rail
GM/RC1500	Code of Practice for EMC Between the Railway and its Neighbourhood
GE/RT8015	Electromagnetic Compatibility between Railway Infrastructure and Trains

Directive 2004/108/EC	Approximation of the Laws of the Member States relating to Electromagnetic Compatibility and repealing Directive 89/336/EEC
SI 2006 No 3418	The Electromagnetic Compatibility Regulations 2006
BS EN 50121-1	Railway Applications. Electromagnetic Compatibility – General
BS EN 50121-2	Railway Applications. Electromagnetic Compatibility – Emission of the Whole Railway System to the Outside World
BS EN 50121-3-1	Railway Applications Electromagnetic Compatibility – Rolling Stock. Train and Complete Vehicle
BS EN 50121-3-2	Railway Applications Electromagnetic Compatibility – Rolling Stock Apparatus
BS EN 50121-4	Railway Applications Electromagnetic Compatibility – Emission and Immunity of the Signalling and Telecommunications Apparatus
BS EN 50121-5	Railway Applications. Electromagnetic Compatibility – Emission and Immunity of Fixed Power Supply Installations and Apparatus
BS EN 12015	Electromagnetic Compatibility. Product Family Standard for Lifts, Escalators and Moving Walks. Emission

BS EN 12016	Electromagnetic Compatibility. Product Family Standard for Lifts, Escalators and Moving Walks. Immunity
BS EN 55015	Limits and Methods of Measurement of Radio Disturbance Characteristics of Electrical Lighting and Similar Equipment
BS EN 61547	Specification for Equipment for General Lighting Purposes. EMC Immunity Requirements
BS EN 55103	Electromagnetic Compatibility. Product Family Standard For Audio, Video, Audio-Visual And Entertainment Lighting Control Apparatus For Professional Use
BS EN 62040	Uninterruptible Power Systems (UPS). Electromagnetic Compatibility (EMC) Requirements
BS EN 60439	Low-Voltage Switchgear and Control Gear Assemblies. Type-Tested and Partially Type-Tested Assemblies
BS EN 61000-6-4	Electromagnetic Compatibility (EMC). Generic Standards. Emission Standard For Industrial Environments
BS EN 61000-6-2	Electromagnetic Compatibility (EMC). Generic Standards. Immunity For Industrial Environments

BS EN 50130	Alarm Systems. Electromagnetic Compatibility. Product Family Standard: Immunity Requirements For Components Of Fire, Intruder And Social Alarm Systems
BS EN 61000-5	Electromagnetic Compatibility (EMC). Installation and Mitigation Guidelines
BS EN 50174	Information Technology. Cabling installation
BS EN 50238	Railway Applications. Compatibility Between Rolling Stock And Train Detection Systems
BS EN 50122	Railway Applications – Fixed Installations – Electrical Safety, Earthing & the Return Circuit.
ITU Directive - Vol. VI	Protection of Telecommunications Lines Against Harmful Effects from Electrical Power and Electrified Railway Lines
Electricity Association	Engineering Recommendation G5/4 Planning Levels for Harmonic Distortion and the Connection of Non-Linear Equipment to Transmission Systems and Distribution Networks in the United Kingdom

Note: References to particular EU Directives and Regulations, Acts of Parliament, Statutory Instruments or Common Law are made only if the subject demands them. Users of engineering standards are bound by all the relevant requirements of the law, regardless of whether or not there is any reference to them in the standards.

4. Abbreviations & Definitions

Abbreviation	Definition
A/m	Amp per meter
AC	Alternating Current
BS	British Standard
CE	Symbol denoting compliance with relevant EU Directives
DC	Direct Current
DLR	Docklands Light Railway
EEA	European Economic Area
EMC	Electromagnetic Compatibility
EN	European Norm
EU	European Union
ESD	Electrostatic Discharge
ITU	International Telecommunications Union
LU	London Underground Ltd
mT	Millitesla (Source: International System Unit)
NR	Network Rail
PFI	Private Finance Initiative
Rms	root mean square
TDMA	Time Division Multiple Access
UK	United Kingdom

Term	Definition
Apparatus	Any finished appliance or combination of appliances made commercially available as a single functional unit, intended for the end user and liable to generate electromagnetic disturbance, or the performance of which is liable to be affected by such disturbance and includes: (a) components or sub-assemblies intended for incorporation into an apparatus by an end-user, which are liable to generate electromagnetic disturbance, or the performance of which is liable to be affected by such disturbance. (b) mobile installations defined as a combination of apparatus and, where applicable, other devices, intended to be moved and operated in a range of locations. Examples include: A computer; power supply, lift controller, rolling stock, a system and assembly of Apparatus
DLR Company Secretary	DLR staff with responsibility for acquisition of general and specific goods and services in accordance with defined commercial arrangements
Equipment	Any apparatus or fixed installation
Electromagnetic Compatibility	The ability of apparatus to function satisfactorily in its electromagnetic environment without introducing intolerable electromagnetic disturbance to other apparatus in that environment.
Electromagnetic Disturbance	Any electromagnetic phenomenon which may degrade the performance of apparatus. An electromagnetic disturbance may include: (i) electromagnetic noise, (ii) an unwanted signal, or (iii) a change in the propagation medium itself.

Fixed installation	A particular combination of several types of apparatus and, where applicable, other devices, which are assembled, installed and intended to be used permanently at a pre-defined location. Examples include: The railway, a station, an airport, telecommunications network and a power station
NR	Network Rail
Supplier	An infrastructure, PFI company, firm, partnership, DLR internal contractor or any other organisation responsible for the provision and maintenance of railway assets for use by DLR
System	A combination of engineering assets placed together or arranged into a regular and connected whole

5. Requirements

5.1 General

5.1.1 The following instructions will ensure that DLR remains in compliance with the EMC Regulations and ultimately the EMC Directive and has a process to address equipment and installation EMC. This process includes the requirements for and implementation of an EMC Control Plan to control EMC activities throughout a project lifecycle, and the demonstration that good EMC engineering practice has been followed throughout the project in the form of EMC assurance document submissions. These may include EMC assessments such as hazard identification and mitigation, risk assessments, gap analyses, certificates of conformity, test and modelling reports and reasoned engineering technical rationale, as appropriate. Independent EMC Notified Body assessment of such submissions is not a statutory requirement. However, depending on the size and complexity of a particular project, it may be appropriate that this becomes a DLR project requirement. Projects may seek guidance on this issue from the DLR Head of Engineering.

5.1.2 Harmonised standards exist for many types of equipment and environments. In addition a set of railway standards (references, BS EN 50121) which apply specifically to the major categories of railway equipment are available. The formulation of an EMC specification for a project may reference published harmonised standards. However it should fundamentally be based on an understanding of the apparatus and systems to be delivered and their operating environment, including DLR and nearby third parties. In some cases compliance with harmonised standards alone will not be sufficient to demonstrate compliance with the EMC Regulations.

5.1.3 It should be noted that the basic requirement to harmonise the electromagnetic environment also applies to in-house developed apparatus and appropriate considerations must therefore be applied in their design, implementation and maintenance.

5.1.4 EMC assessments shall also address any credible adverse apparatus EMC performance effects caused by:

- Manufacturing tolerances;
- Ageing;
- Usage;
- Corrosion.

Note – The issues to be addressed could include the following examples:

1. Manufacturing tolerances – Excessive variations in filter attenuation or surge device clamping performance due to production tolerances;
2. Ageing – Degradation in the shielding effectiveness of a cabinet door gasket due to mechanical shock, vibration, an excessive number of compressions, temperature, humidity or solar radiation.
3. Ageing – Failure of surge protection devices due to an excessive number of voltage clamping operations;
4. Usage – The circumvention of apparatus shielding performance due to damage, an access door or panel not being correctly closed or secured;
5. Usage – The loosening of a 360° cable screen termination due to vibration;
6. Corrosion – A reduction in filtering performance due to a corroded/high impedance bonding connection;
7. Corrosion – A reduction in cable containment shielding effectiveness due to corroded metalwork.

Suppliers should address these issues during the design process, by the proper selection of components & materials etc, and include any necessary mitigation during the periodic inspection and maintenance cycles.

Note that compliance with the standards of other infrastructure operators, especially London Underground and Network Rail (NR), may also be required, according to asset type and location.

5.1.5 The EMC Control Plan or specifications prepared from this standard shall clearly define the methods by which the requirements of the standard are to be verified, taking account of any measurable criteria and tests described herein.

5.2 New Equipment

Apparatus

5.2.1 Most apparatus supplied is required to be CE marked for the environment in which it is to be used, confirming that all relevant directives have been met. Documentary evidence shall be provided in all cases, whether or not any product is CE marked, to confirm that it is suitable for use on the DLR.

> **Note** – Apparatus that is not included within the scope of EN 50121, or by any other DLR EMC requirement, which has been tested and certified to a current, generic industrial or industrial based emission and immunity product standards may be considered compliant, subject to agreement with the DLR Head of Engineering.

Fixed Installation

5.2.2 The Directive now implements a separate regulatory regime for Fixed Installations. Suppliers shall develop and implement EMC design, installation & maintenance processes and policies to ensure that a Fixed Installation is installed & maintained in such a way to achieve EMC and compliance with the Directive. Documentary evidence shall be provided to DLR in all cases but no single CE marking is required for the whole Fixed Installation.

5.3 Maintenance

5.3.1 An assessment shall be made as to whether the electromagnetic characteristics of apparatus are likely to be affected by maintenance operations. If this is the case, e.g. if like-for-like replacement is not possible, the modified equipment shall be treated as for new equipment. The maintainer shall carry out such assessments or refer them to other appropriately qualified personnel and shall retain evidence of these assessments.

5.3.2 For like for like replacement a formal EMC assessment is required in accordance with 5.3.1 unless the electrical and electromagnetic properties of the replacement are identical with those of the substituted item.

5.4 Modifications

5.4.1 Project Managers, Project Engineers and Suppliers shall ensure that the electromagnetic characteristics of apparatus, systems & installations are assessed within the project. If any modification might affect these characteristics, a formal EMC assessment shall be required.

5.5 Advice

5.5.1 The DLR Head of Engineering is responsible for advice on specific EMC issues.

5.6 DLR Specific Needs

5.6.1 Some specific DLR needs are not covered in detail in BS EN 50121. Project Managers, Project Engineers and Suppliers or their nominated representative(s) shall therefore ensure that these aspects are addressed within a project:

Signalling Compatibility

5.6.2 Signalling systems shall be designed and installed such that they are electromagnetically compatible in their operating environment and comply with this document and the UK EMC Regulations. Signalling projects shall undergo thorough assessment of all EMC hazards and risks and close all hazards out before the system is brought into use.

5.6.3 For projects involving signalling technology which has not been previously used on DLR independent EMC Notified Body assessment of EMC assurance evidence is required. Projects may seek guidance from the DLR Head of Engineering on this matter.

Psophometric Noise

5.6.4 The basic principles are covered in BS EN 50121-3-1 and International Telecommunications Union (ITU) Directives “Protection of telecommunications lines against harmful effects from electrical power and electrified railway lines” – Vol. VI Danger and Disturbance, which stipulates a maximum psophometric interference level of 1 mV for analogue telecommunications.

DC and Low Frequency Magnetic Fields

5.6.5 Such requirements, principally in the context of rolling stock, are to minimise risk to passengers with active implantable medical devices such as pacemakers. Rolling stock suppliers shall demonstrate that they have assessed and mitigated this risk through design and test as appropriate.

Protection of DLR Radio Services

5.6.6 The Rolling stock radiated emissions, contained within BS EN 50121-3-1, have been chosen by the European standards committee to protect radio services external to the railway. These limits may not adequately protect the operation of internal DLR radio services and require adjustment in accordance with the performance properties and technical characteristics of such Equipment. Within the DLR there are a number of radio services located within close proximity to rolling stock and the threat from radiated emissions needs to be fully considered during the supplier's EMC assessment. These radio services are likely to include both licensed and license exempt Equipment with the latter being particularly susceptible because of their low levels of permitted radiated power levels and corresponding need for a very high signal to noise protection ratio.

Protection from Mobile Phone and Radio Handsets

5.6.7 The wide scale and ever increasing usage of mobile handsets by public and staff represents an increased operational risk to DLR apparatus. This additional risk occurs because the operational radiated electric field threat level can easily exceed the apparatus certification levels, which are generally based on the unrealistic expectation that a protection or separation distance is achieved between the mobile phone and any apparatus, contained within many European & International EMC standards. The supplier's EMC Control Plan shall discuss and address these risks.

Induced accessible and touch voltages

5.6.8 DLR requirements for apparatus bonded to the running rails and for the control of rail potentials are given in DLR-ENG-STD-ES-604 §6.1.

Electrostatic Discharge

5.6.9 Apparatus shall be capable of withstanding such effects in accordance with BS EN61000-4-2 requirements (4 kV contact, 8 kV air discharge).

5.7 Evidence of Compliance

5.7.1 Compliance with the requirements of this standard shall be demonstrated to DLR by each party contracted to DLR. This may be by the provision of, EMC Control Plans, EMC test evidence, EMC studies, EMC HAZIDs, EMC GAP Analyses, EMC Test Plans and Reports, and EMC Project Files etc as appropriate depending on the complexity of the project. The EMC Compliance documentation should detail the following items:

- identify the apparatus, system or installation and its EMC characteristics;
- identify and understand the possible EMC risks;
- show how the apparatus, systems & installations shall meet the requirements of the EMC Regulations Statutory Instrument 2006 No 3418 and this standard;
- address life cycle issues in order to ensure on-going EMC during the life of the apparatus, system or installation;
- document good EMC engineering practice to be used;
- clearly define the methods by which the requirements of the standard are to be verified, taking account of any measurable criteria and tests described herein; and
- identify an ‘EMC’ Responsible Person.

5.7.2 Additionally DLR may audit compliance as part of its surveillance regime.

5.7.3 Independent EMC Notified Body assessment of EMC assurance evidence submissions is not a statutory requirement. However, it may be appropriate depending on the size and complexity of a particular project that it is a DLR project or contract requirement. Projects may seek guidance on this issue from the DLR Head of Engineering.

5.8 EMC Deliverables

5.8.1 The EMC deliverables will depend on the size and complexity of each project and should be agreed by the contractor with DLR at an early stage. An outline of deliverables against project stage is given below.

AiP

EMC Control Plan

AfC/AoD

EMC HAZID & Risk Log

EMC Design Closeout Report / Check Certificate

AfT

EMC Equipment Assurance Evidence

EMC Gap Analysis (if appropriate)

EMC Test Plan(s) (if appropriate)

AfO

Completed EMC installation inspection evidence (for example ITPs)

EMC Test Report(s) (if appropriate)

Completed EMC Project File

EMC Control Plan

5.8.2 The Main Contractor shall produce an EMC Control Plan to demonstrate how the requirements of this document and the UK statutory requirements will be met for the whole project. The EMC Control Plan shall identify which standards, criteria and methods are intended to be used for the assessment of EMC as well as defining project roles and responsibilities for EMC. Contractors shall also take cognisance of the requirements of integrated system testing that will be undertaken as part of the system handover. For some projects EMC site testing may be required. However, in many instances on site tests will be limited to visual inspection of the installation and completion of site verification checklists (via Inspection Test Plans) as well as confirmation that individual equipment complies with the EMC Regulations. The project EMC Control Plan shall be submitted at AiP and shall be updated by the Contractor at the start of detailed design and throughout the project lifecycle as necessary.

EMC HAZID & Risk Log

5.8.3 Contractors shall identify all potential interfaces between their equipment/systems and existing equipment/systems located within the DLR environment and with third parties such as LU & NR etc [NB this is not an exhaustive list] and undertake an EMC HAZID. Risks should be recorded, tracked and closed out throughout the project. It may not be possible to close out all risks by AfC and some elements of the risk log may need to be closed out by during test & commissioning (for example through the successful execution of the ITPs).

EMC Design Closeout Report / Check Certificate

5.8.4 The Designer shall provide confirmation that the completed design is EMC & complies with the project EMC Control Plan through an EMC Design Closeout Report or Check Certificate. This shall outline the EMC activities which have taken place during the design, highlight any particular EMC issues which occurred during the design and detail how these were resolved and be authorised by the Designer's EMC Engineer. For multidiscipline projects evidence of effective integration of the designs will be required. The document shall be available at AfC.

Inspection Test Plans (ITP)

5.8.5 Projects shall take account of EMC in their drafting of ITPs and ensure that checks are included to verify the EMC integrity of installations. These may include checking correct bonding, cable separation, equipment location in relation to significant threats etc.

EMC Project File

5.8.6 The Contractor shall begin the EMC Project File at the design stage and populate it with EMC design evidence as appropriate. This will include for example, the EMC Control Plan, EMC Risk Log, equipment assurance evidence (certificates/test reports etc), any necessary gap analyses and the output of any EMC studies undertaken. During installation, test and commissioning the Contractor shall complete the EMC Project File adding such things as test certificates/reports, completed ITPs etc and the completed EMC Project File shall form part of the handover documentation at AfO.

5.9 Lifts, Escalators & Moving Walks

5.9.1 This section contains product specific test and documentation requirements for Lifts, escalators & moving walks and are largely consistent with those of London Underground (S1222)..

5.9.2 Lifts, Escalators and Moving Walks shall be tested and certified in accordance with the current versions of BS EN 12015 and BS EN 12016 by an accredited and independent test house.

Note: Lifts, Escalators & Moving Walks with a rated input current in excess of 100A may need to comply with additional EMC requirements in consultation with the DLR Head of Engineering.

Modes of Operation

5.9.3 The EMC tests shall be conducted on the apparatus in the operational mode(s) producing the highest emissions and susceptibility consistent with the normal operation & configuration.

Note: The following modes shall be considered to determine the worse case mode of operation:

a) Soft start;

b) Regeneration braking;

c) Resistive braking;

d) Standby;

e) Maintenance;

f) Full load;

g) None or minimal load;

h) Emergency Stop.

Harmonics

5.9.4 The Product Standard BS EN 12015 contains requirements for harmonic distortion. Projects & Contractors shall ensure that the requirements of the

Electricity Association Engineering Recommendation G5/4 are met for sites where L&E asset works are undertaken.

Immunity above and beyond the requirements of BS EN 12016

5.9.5 The apparatus shall perform as per the test requirements set out below. Any higher test levels shall take precedence over those defined in the BS EN 12016, whose scope does not include the more severe environments such as railways or metros, and have therefore been included to more closely reflect the DLR EMC environment.

5.9.6 During immunity testing the fault indicators and health monitoring instrumentation apparatus shall be tested as safety circuits.

5.9.7 In the event of detecting any susceptibilities, the failure conditions and the assembly of apparatus's performance shall be documented to support any safety, reliability assessments / investigations. The stimulus threshold and frequency at which the assembly of apparatus fails shall also be determined and documented.

Note: The final acceptance of any immunity failures, below the DLR requirements, shall be at the discretion of the DLR Head of Engineering.

5.9.8 Radiated immunity testing (Tests 1, 2 and 3 only) shall be performed with apparatus cabinet doors or panels open to simulate the use of mobile transmitters during maintenance or fault rectification. The assembly of apparatus must fail-safe and not be damaged should any susceptibilities occur.

Table 1 – Immunity test requirements

Test	Environmental Phenomena	Test Values		Circuit Performance Criteria Note 8	
		All circuits	Safety circuits	All circuits	Safety circuits
1	TETRA radiated Immunity 382MHz – 385MHz Note 1	10V/m (rms) TDMA	30V/m (rms) TDMA	A	D
2	TETRA radiated Immunity 392MHz – 395MHz Note 1	10V/m (rms) TDMA	30V/m (rms) TDMA	A	D
3	DLR Radio 453MHz – 463MHz	10V/m (rms) TDMA	30V/m (rms) TDMA	A	D
4	50Hz power frequency magnetic field Notes 2 & 7	100A/m for 5s	100A/m for 5s	A	A
5	Pulsed magnetic field Notes 2, 3, 5 & 7	300 A/m	300 A/m	A	A
6	Conducted Immunity 0Hz – 150kHz Notes 4 & 7	10V	10V	A	A
7	Fast transient burst Notes 5, 6 & 7	1kV	2kV	B	D
8	Power surge – Line to earth Notes 5 & 7	2kV	2kV	B	D

9	Power surge – Line to line Notes 5 & 7	1kV	1kV	B	D

Notes

1. Refer to BS EN 61000-4-3 or an equivalent;
2. BS EN 61000-4-8: Only applicable to Apparatus that contains magnetically sensitive components;
3. BS EN 61000-4-9; Applicable to Apparatus installed <3m from the track;
4. BS EN 61000-4-16: Applicable to cables >20m in length;
5. The product standard immunity levels may be acceptable if it can be shown that the installation technique can mitigate the disturbance;
6. BS EN 61000-4-4: – Applicable to port interfaces for monitoring and remote alarm Apparatus with cables greater than 3m and crossing Equipment boundaries;
7. If any of the Equipment design aspects are uncertain (i.e. the use of magnetically sensitive components, distance from the track or cable lengths etc.) then this test should be applied to avoid any retrospective testing or analysis;
8. Refer to BS EN 12016:2004 + A1 2008 - EMC - Product family standard for lifts escalators and moving walks - Immunity.

L&E Documentation requirements

5.9.9 The supplier shall provide and maintain adequate technical documentation to demonstrate EMC.

Note: Documentation should include, but not be limited to, the following:

a) EMC Control Plan;

b) On completion of EMC testing or compilation of evidence that EMC tests have previously been undertaken, a technical file, including test report(s) and declaration of conformity with the EMC Directive shall be provided;

c) Installation policy & practices (e.g. cable separation, cable screen termination methods, containment; corrosion control, earthing & bonding and any additional filtering or other requirements needed to achieve EMC);

d) Electrostatic device handling precautions;

e) Installation check lists & inspections reports to confirm the above have been properly implemented;

f) EMC maintenance requirements;

g) Declaration of Conformities;

h) Site inspection and test reports confirming that all relevant equipment associated with EMC compliance has been installed in accordance with the requirements of this standard, and “good engineering practices” as required by The Electromagnetic Compatibility Regulations 2006.

i) A detailed justification explaining why any previously undertaken EMC testing is equivalent to the requirements contained this standard;

j) Where previous EMC testing of the Apparatus was compliant to the DLR requirements at the time or is considered to be equivalent to the current test requirements, the supplier should consult DLR to determine if the evidence is still acceptable.

5.10 EMC Non-compliances

5.10.1 There may be instances where evidence for demonstrating apparatus' compliance with the DLR EMC requirements is not available. Business Critical Process DLR-IMS-GENR-BCP-00009, states the procedure to be followed where this is the case.

5.10.2 Where an EMC non-compliance has been identified (e.g. through inspection, audits, incident investigations or other monitoring activities) the Project Manager, Project Engineer, or Supplier shall take all reasonable steps to rectify the non-compliance as soon as practicable, and before entering the asset into service.

5.10.3 As per DLR-IMS-GENR-BCP-00009, when having been made aware of a non-compliant asset, the Project Manager, Project Engineer, or Supplier shall arrange for a suitably qualified and experienced person to undertake an immediate (real time) assessment of the safety risk associated with the non-compliance.

6. Responsibilities

6.1.1 The DLR Head of Engineering is the EMC Responsible Person for all DLR assets. The EMC Responsible Person shall be, by virtue of their control of the fixed installation, able to determine that the configuration of the installation is such that when used it complies with the essential requirements; and shall hold all EMC data in relation to DLR assets at the disposal of the enforcement authority for inspection purposes for as long as the asset is in operation.

6.1.2 The DLR Head of Projects or (responsible person to be agreed with the Head of Engineering where the change is not a project) shall be responsible for ensuring that a co-ordinated programme of audit and inspection is implemented to demonstrate that compliance with this and other related standards are achieved.

6.1.3 DLR Company Secretary shall be responsible for incorporating the requirements of this engineering standard in any contract to which it is relevant and shall stipulate that a programme of audits is implemented by the contractor to ensure that these requirements are met. This programme and its results shall be available for verification by the DLR Head of Projects.

6.1.4 Procurement Managers shall be responsible for ensuring that apparatus, systems and installations purchased meets the requirements of this standard and is marked accordingly.

6.1.5 Project Engineers shall be responsible for ensuring that the apparatus, systems and installations comprising project deliverables, whether newly commissioned or modified, meet the requirements of statutory requirements and this standard and are marked accordingly.

6.1.6 Project Managers shall have responsibility for ensuring that the total apparatus, system or installation delivered or modified by a project complies with statutory requirements and the requirements of this standard and are marked accordingly.

6.1.7 Suppliers shall, by enforcing appropriate requirements on manufacturers, designers and other parties in their supply chain, be responsible for achieving and demonstrating full compliance with the EMC Directive, this standard and marking apparatus, systems and installations accordingly. Suppliers shall ensure sufficient compliance evidence/controlled information is made available and is maintained to show that good engineering practices have been applied.

7. Supporting Information

7.1 Background

7.1.1 The original EMC Directive was first transposed into UK law by the EMC Regulations 1992 and, following a transition period, its provisions applied fully from 1st January, 1996. Directive 2004/108/EC was transposed into UK Law by the EMC Regulations 2006 (SI 2006 No. 3418), and largely came into force on the 20 July 2007. Following a transition period for some aspects, its provisions have applied fully from the 20 July 2009. The EMC Directive now implements a separate regime for a Fixed Installation but it does not include safety related requirements, which are addressed by other parts of this document and other DLR standards.

7.1.2 The EMC Directive applies to most electrical and electronic equipment and one of its main purposes is to remove barriers to trade within the European Economic Area (EEA). An additional key purpose is to ensure that any apparatus-generated electromagnetic disturbances do not exceed a level above which radio and telecommunications apparatus and other apparatus cannot operate as intended, and that the apparatus itself has an adequate level of immunity to an electromagnetic disturbance.

7.1.3 This standard aims to ensure that DLR and its suppliers meet these legislative requirements. The provisions of this standard apply to DLR procurement agents, suppliers contracted under any subsequent specifications and any technical and maintenance staff within DLR.

7.2 Safety Considerations

7.2.1 Although the EMC Directive is not explicitly safety related, its fundamental aim of promoting EMC in apparatus, systems and installations will have benefits in terms of both safety and reliability. In the DLR context the principal concern is the avoidance of unsafe conditions from interference effects in signalling and other control equipment. Thus, EMC is an issue which is specifically addressed in internal approval processes, often for submissions to London Underground, Network Rail, and similar bodies.